BridgeIt

Project Overview (Product Requirements Document) · Draft v0.3 · 22 October 2026

A phone game that teaches students how to connect sentences with coordinating conjunctions

> **How to read this document.** It describes what the BridgeIt app will do and how it will look, so everyone agrees before building starts. Items marked **To confirm** still need your decision. They are all collected as questions in Section 10. Items marked *Our proposal* are suggestions from the development team that you can accept or change.

1. What is BridgeIt?

BridgeIt is a game for Android phones. It teaches students the seven coordinating conjunctions, remembered as **FANBOYS**: *for, and, nor, but, or, yet, so*.

In the game, two parts of a sentence stand on opposite sides of a river. The student builds a bridge between them by choosing the right conjunction. When the answer is right, the bridge is completed and a character's vehicle drives across toward its destination.

The game is used in class as part of a study led by teacher-researchers. At the end of each level, the teacher gives the student a code that opens the next level. This way, the teacher decides when students move on.

2. What students will learn

After playing all three levels, the student should be able to:

1. **Remember** the seven FANBOYS conjunctions and what each one does.
2. **Use** them to join two complete sentences correctly.
3. **Understand** their meaning well enough to pick a word with the same meaning (for example, *but* and *however*).

Not included in this version

- Other kinds of conjunctions, such as *because, although* and *while*. The earlier web prototype had some of these. This version focuses on FANBOYS only.
- Student accounts, logins or a leaderboard.
- A version for iPhone or for web browsers.

3. Who will use it

Who	What they need
Students	A simple, encouraging game. Clear feedback when an answer is wrong, and a chance to try again. Grade level: **To confirm**
Teacher-researchers	Control over when students move to the next level, using codes. Possibly a record of each student's results: **To confirm**

4. Where it runs

- **Device:** Android phones, held upright.
- **Delivery:** an Android app file (APK) that can be installed on the students' phones.
- **Language:** English only.
- *Our proposal:* the app works without an internet connection, and students do not need to sign in.
- *Our proposal:* there is a button to turn the sound on and off.

5. How the game flows

1. **Welcome screen.** The character greets the player ("Hi, Bridge Builder!") and explains the goal.
2. **Level menu.** Shows Levels 1, 2 and 3. Levels 2 and 3 stay locked until the teacher gives the code.
3. **Levels 1 to 3**, described in Section 6.
4. **Finish.** After Level 3, the vehicle reaches the character's destination.

6. The three levels

Level 1 – Conjunction Song

- The student listens to a song while the lyrics are shown on screen. The song explains what conjunctions are and teaches FANBOYS. You will provide the song recording later. Until then, the lyrics are shown on their own.
- The student then fills in the missing conjunctions in the lyrics by tapping the right word.
- The student can try again as many times as needed.
- When the activity is done, the teacher gives **code 1**, which opens Level 2.

Level 2 – Building the Bridge

- Two complete sentences sit on either side of a broken bridge (for example, "I'm very tired" and "I will sleep."). The seven FANBOYS words are shown below as bridge pieces. The student drags the right piece into the gap.

- **Right answer:** the bridge is completed and the vehicle crosses.
- **Wrong answer:** the bridge cracks, and a short hint explains why that word does not fit. The student tries again. Number of tries allowed before the bridge breaks: **To confirm**
- Number of sentences in this level: **To confirm**. Whether a sentence can have more than one right answer: **To confirm**
- When all sentences are done, the teacher gives **code 2**, which opens Level 3.

Level 3 – Strengthening the Bridge

- The student first completes the bridge, the same way as in Level 2.
- Then the student strengthens the bridge by choosing a word or phrase with the same meaning as the conjunction they used (for example, *but* → *however*, *so* → *therefore*).
- Wrong choices crack the bridge and show a hint, the same as in Level 2.
- When the level is done, the bridge is reinforced and the vehicle reaches its final destination.

7. What the app will do

A checklist of everything the finished app must do:

- Show a welcome screen with Back and Next buttons.
- Show a level menu where Levels 2 and 3 open only with the teacher's code.
- Remember which levels a student has opened, even after the app is closed.
- Level 1: play the song with lyrics, then run the fill-in-the-blanks activity with unlimited tries.
- Levels 2 and 3: let the student drag conjunction pieces into the gap in the bridge.
- Right answer: complete the bridge and move the vehicle forward. Wrong answer: crack the bridge and show a hint.
- Level 3: ask for a word with the same meaning after the bridge is built.
- Let the team change the sentences, answers, hints, same-meaning words, number of tries and codes easily, without rebuilding the whole app.
- Let the student turn the sound on and off.

8. Learning content (draft)

The meanings below follow the song in your materials. Please confirm or correct them.

Conjunction	What it does	Same-meaning word used in Level 3
for	Gives a reason	**To confirm**
and	Adds an idea	**To confirm**
nor	Adds a second "not" idea	**To confirm**

Conjunction	What it does	Same-meaning word used in Level 3
but	Shows a contrast	however
or	Gives a choice	**To confirm**
yet	Shows a surprising contrast	**To confirm**
so	Shows a result	therefore

9. Look and feel

Page 2 of your Canva design has the BridgeIt logo: a black line drawing of a lotus. No colors were given, so we prepared a **proposal** for you to review.

Proposed colored logo



Your original drawing is kept exactly as it is. We only added color: pink petals with deep-rose outlines, green swirls, a golden flame and teal dots.

Proposed app colors

The app's colors are taken from the colored logo and from the game's scenery (sky, river, cliffs and bridge).

Color	Where it is used	Color code
	Main buttons, river	#0E7490
	Highlights, taken from the lotus	#EC4899
	Conjunction pieces, logo outlines	#9D174D
	Screen background (sky)	#E0F2FE
	Text, "Next" and "Start" buttons	#1E293B
	Bridge planks	#B45309
	Cliffs and grass, completed levels	#15803D
	Vehicle, hints	#F59E0B
	Right answers	#22C55E

Color	Where it is used	Color code
	Wrong answers, bridge cracks	#DC2626

Right and wrong answers are also shown with an icon, words and movement, not just color, so students who have trouble telling colors apart are not left out.

Rough screen designs

Rough drawings of the welcome screen, level menu, Level 1, Level 2 and Level 3 are sent with this document as a separate PDF (*BridgeIt-Wireframes.pdf*). They show what goes on each screen and how the colors are used. The character, vehicle and scenery are stand-ins, not final artwork.

> **If you prefer a different look.** The colored logo, the colors and the screen designs are proposals only. You may send your preferred colored logo or design instead. If you do, the colors from your version will be used for the whole app, and the screen designs will be updated to match.

- Style of the character, vehicle, bridge and background: **To confirm**
- Sound effects besides the Level 1 song: **To confirm**

10. Questions for you

#	Question
1	What grade level are the students?
2	Will students play with a teacher guiding the class, on their own, or both?
3	Who types in the codes that open Levels 2 and 3, and should the codes be fixed or chosen by the teacher?
4	How many sentences should Levels 2 and 3 have?
5	How many wrong answers can a student make before the bridge breaks? When it breaks, should the student redo that sentence or the whole level?
6	Can a sentence have more than one right conjunction?
7	Do the researchers need the app to save each student's results (number of tries, scores, time taken)? This would need an internet connection and an online storage service, which adds work.
8	Which same-meaning word should Level 3 use for each conjunction (Section 8)?
9	Will the BridgeIt logo be used in the app (welcome screen, app icon)?
10	Do you approve the proposed colored logo and app colors (Section 9)? If not, please send your own colored logo or design. Its colors will be used for the whole app.
11	Do the rough screen designs (BridgeIt-Wireframes.pdf) match what you have in mind? What should change?

#	Question
12	What style should the character, vehicle, bridge and background have?
13	Should the game have sound effects besides the song?
14	When will the song recording be ready?

11. Attachments and references

- Your Canva design "Intervention Idea" (BridgeIt screens and song lyrics).
- Proposed colored logo: *BridgeIt-logo-color.svg* and *BridgeIt-logo-color.png*.
- Rough screen designs: *BridgeIt-Wireframes.pdf*.
- Earlier prototype: *https://conjbridge.vercel.app/*. Used only as a reference for the bridge idea. Its questions and extra conjunctions are not part of this project.